

CHAPTER 4

Diodes

Outline

- 4.1 The ideal diode
- 4.2 Terminal characteristics of junction diodes
- 4.3 Modeling the diode forward characteristics
- 4.5 Rectifier circuits
- 4.6 Limiting and clamping circuits

4.1 The ideal diode

- 4.1.1 Current-voltage characteristic (non-linear/piecewise linear)
 - Reverse bias: open circuit/cut off, $v < 0, i = 0$
 - Forward bias: short circuit/turn on, $v = 0, i > 0$

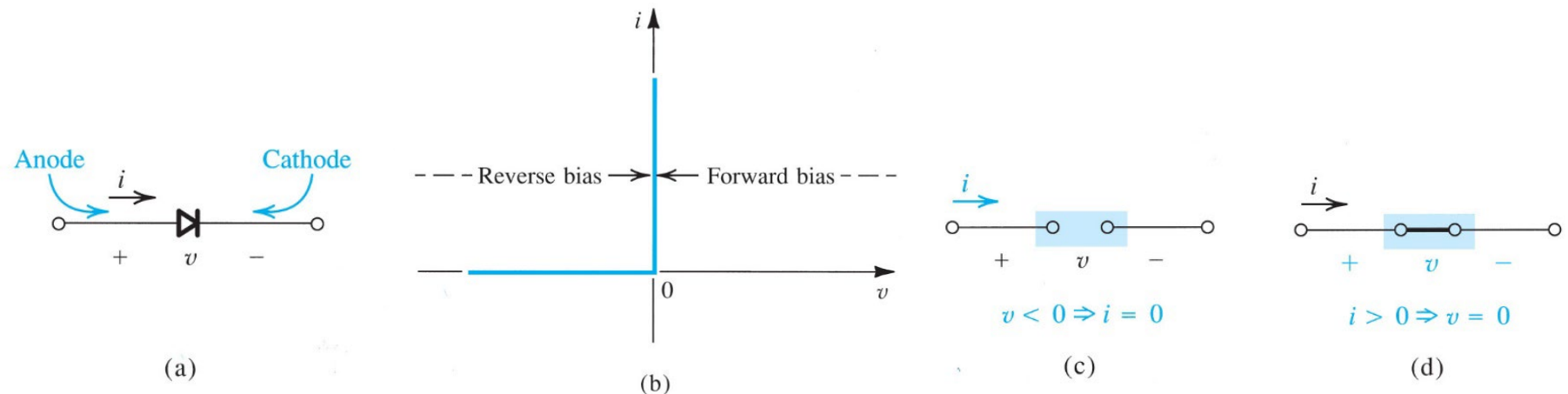


Figure 4.1 The ideal diode: (a) diode circuit symbol; (b) i - v characteristic; (c) equivalent circuit in the reverse direction; (d) equivalent circuit in the forward direction.

- Voltage dropped across the diode (v) and the current flows through the diode (i)
 - Fig. 4.2(a): conducting diode/short circuit, $v = 0, i = \frac{v}{R} = \frac{10V}{1k\Omega} = 10mA$
 - Fig. 4.2(b): cut off/open circuit, $v = 10V, i = 0mA$

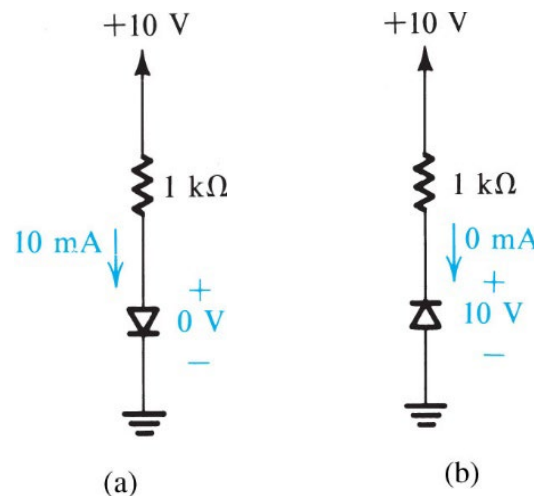


Figure 4.2 The two modes of operation of ideal diodes and the use of an external circuit to limit (a) the forward current and (b) the reverse voltage.

- 4.1.2 A simple application: the rectifier
 - Can be used to generate dc from ac

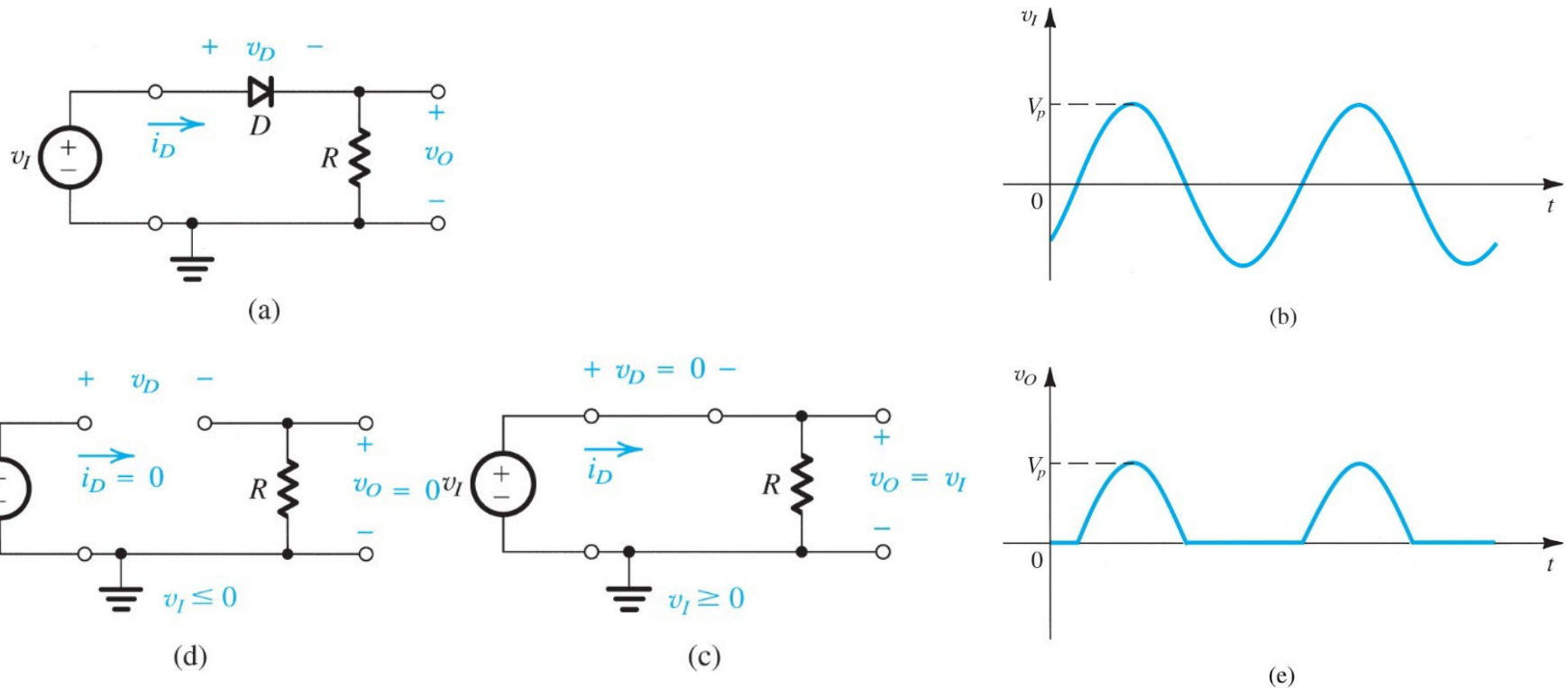


Figure 4.3 (a) Rectifier circuit. (b) Input waveform (c) Equivalent circuit when $v_I \geq 0$. (d) Equivalent circuit when $v_I \leq 0$. (e) Output waveform

4.2 Terminal characteristics of junction diodes

The characteristic curve consists of 3 distinct regions.

- Forward-bias: $v > 0$;
- Reverse-bias: $v < 0$;
- The breakdown region: $v < -V_{ZK}$;

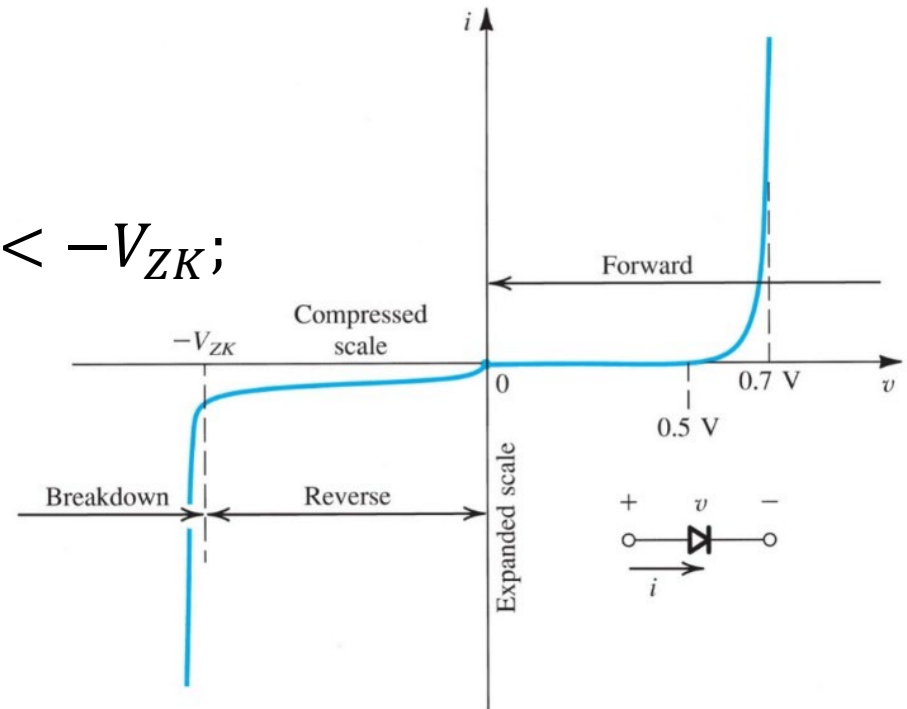
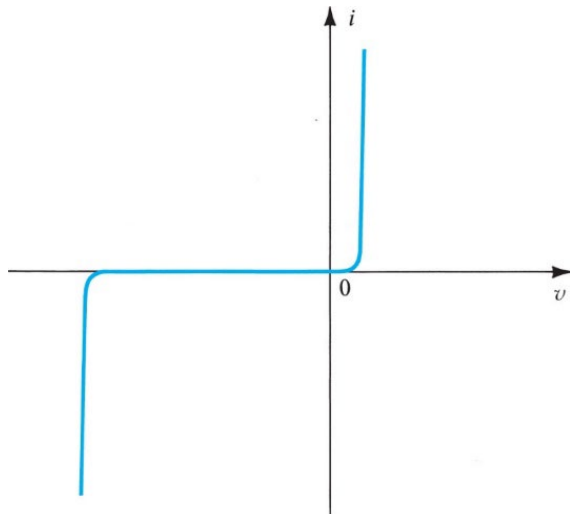


Figure 4.8 The diode i - v relationship with some scales

Figure 4.7 The i - v characteristic of a silicon junction diode, expanded and others compressed in order to reveal details.

- 4.2.1 The forward-bias region

$$i = I_s e^{v/v_T}$$

- I_s : saturation current/scale current, very strong function of temperature
- v_T : thermal voltage, $\approx 25mV$ at room temperature

or $v = v_T \ln \frac{i}{I_s}$, where $\ln$ is the natural logarithm.

- Example 4.3

A silicon diode said to be 1-mA device displays a forward voltage of 0.7-V at a current of 1mA. Evaluate the junction scaling constant I_s .

What scaling constants I_s would apply for a 1-A diode of the same manufacture that conducts 1-A at 0.7-V.

(a) $i = I_s e^{v/v_T}$, so $I_s = i e^{-v/v_T}$,

$$I_s = 1mA \times e^{-0.7V/25mV} = 6.9 \times 10^{-16}A$$

(b) $I_s = 1A \times e^{-0.7V/25mV} = 6.9 \times 10^{-13}A$

• The forward-bias region

If $I_1 = I_s e^{v_1/v_T}$, $I_2 = I_s e^{v_2/v_T}$,

$$\frac{I_1}{I_2} = e^{(v_1 - v_2)/v_T} \quad \text{or} \quad v_1 - v_2 = v_T \ln \frac{I_1}{I_2} \quad \text{or}$$

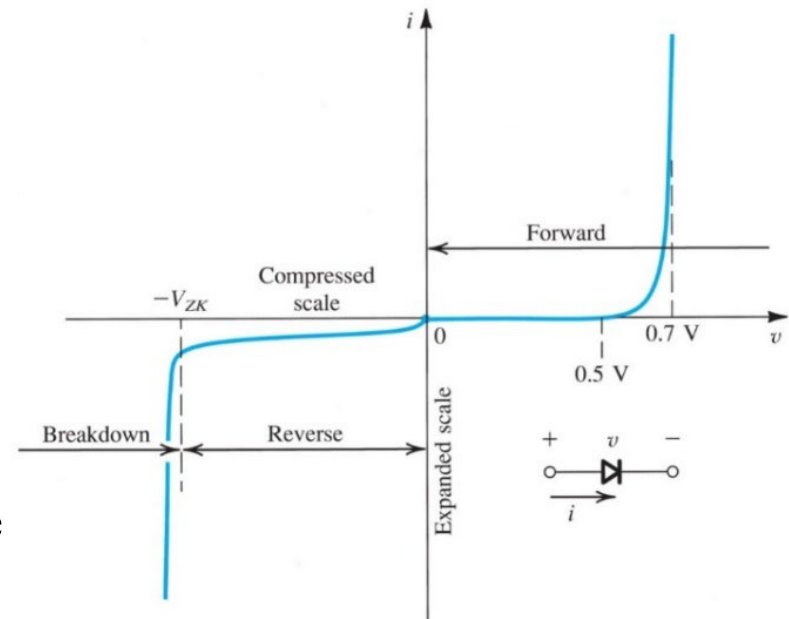
$$v_1 - v_2 = 2.3 v_T \lg \frac{I_1}{I_2}, \quad \lg \text{ is base-10 logarithms}$$

$$\ln x \sim \lg x \Rightarrow y = \lg x \Rightarrow x = 10^y \\ \Rightarrow \ln x = y \times \ln 10 = 2.3y = 2.3 \lg x$$

- **Conclusion #1:** for a decade current change, the diode voltage changes by $2.3v_T$, which is approximately 60 mV. So

$$v_1 - v_2 = 0.06 \times \lg \frac{I_1}{I_2}$$

- **Conclusion #2:**
 - $< 0.5\text{V}$ (**cut-in**), the current is negligibly small
 - $0.6 \sim 0.8\text{V}$, rapid increase of I
 - A simple model for the diode – a conducting diode has approximate a **0.7-V drop across** it.



• Exercise 4.6

Find the change in diode voltage if the current changes from 0.1mA to 10mA.

$$v_1 - v_2 = 2.3v_T \lg \frac{I_1}{I_2} = 2.3 \times 25mV \times \lg \frac{10mA}{0.1mA} = 115mV$$

Or approximately, $\approx 2 \times 60mV = 120mV$

• Exercise 4.7

A silicon junction diode has $v = 0.7V$ at $i = 1mA$. Find the voltage drop at $i = 0.1mA$ and $i = 10mA$

$$v_1 - v_2 = 0.025 \ln \frac{I_1}{I_2}, \text{ so } v_1 = v_2 + 0.025 \ln \frac{I_1}{I_2} = 0.7V + 0.025 \times \ln \frac{I_1}{1}$$

$$\text{For } i = 0.1mA, v_1 = 0.7V + 0.025 \times \ln \frac{0.1}{1} = 0.64V$$

$$\text{For } i = 10mA, v_1 = 0.7V + 0.025 \times \ln \frac{10}{1} = 0.76V$$

Or approximately, $v_1 = 0.7V + 0.06 \times \lg \frac{0.1}{1} = 0.64V$ for $i = 0.1mA$

and $v_1 = 0.7V + 0.06 \times \lg \frac{10}{1} = 0.76V$ for $i = 10mA$

- 4.2.2 The reverse-bias region

$i \approx -I_s$, I_s saturation current

- 4.2.3 The breakdown region

Breakdown voltage: V_{ZK}

where the subscript Z stands for zener and K denotes knee

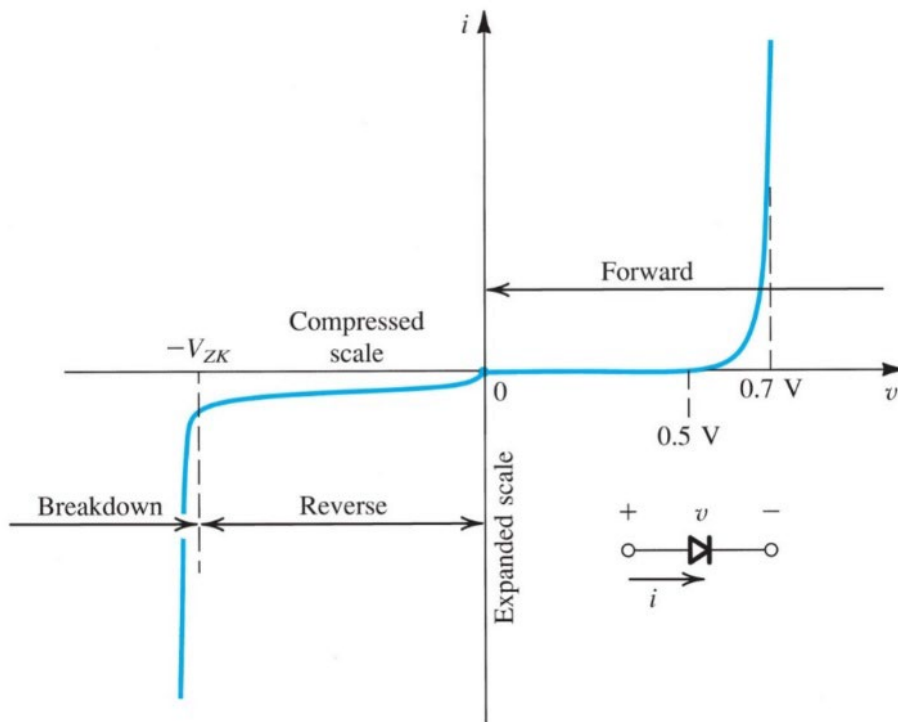


Figure 4.8 The diode i - v relationship with some scales expanded and others compressed in order to reveal details.

4.3 Modeling the diode forward characteristics

- 4.3.1 The exponential model

$$I_D = I_s e^{V_D/V_T} \quad - \textcircled{1}$$

$$I_D = \frac{V_{DD} - V_D}{R} \quad - \textcircled{2}$$

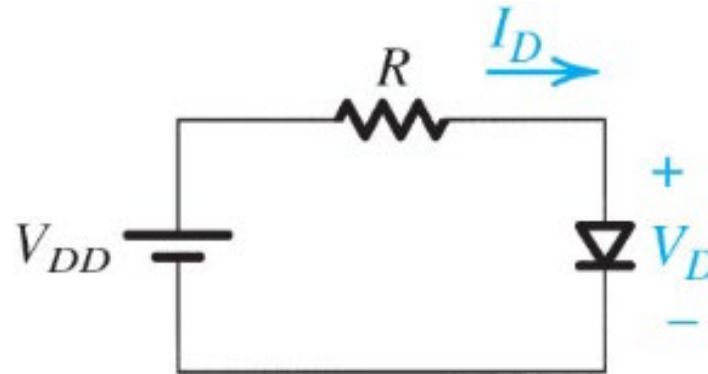


Figure 4.10 A simple circuit used to illustrate the analysis of circuits in which the diode is forward conducting.

• Exercise 4.11

Design the circuit in Fig. 4.11 to provide an output voltage of 2.4V. Assume that the diodes available have 0.7-V drop at 1mA.

$$I = I_s e^{v/v_T}, \text{ so } I_s = I e^{-v/v_T},$$

$$I_s = 1\text{mA} \times e^{-0.7\text{V}/25\text{mV}}$$

$$\text{When } v = \frac{2.4\text{V}}{3} = 0.8\text{V}$$

$$I = 1\text{mA} \times e^{-0.7\text{V}/25\text{mV}} e^{0.8\text{V}/25\text{mV}} = 54.6\text{mA}$$

$$R = \frac{10\text{V} - 2.4\text{V}}{54.6\text{mA}} = 139\Omega$$

Alternative way:

$$0.8 - 0.7 = 0.025 \times \ln \frac{I_1}{1\text{mA}}, \text{ so } I_1 = 54.6\text{mA}$$

$$\text{Or } 0.8 - 0.7 = 0.06 \times \lg \frac{I_1}{1\text{mA}}, \text{ so } I_1 = 46.4\text{mA}$$

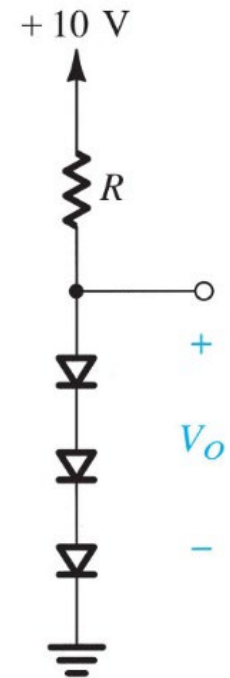


Figure E4.11

• 4.3.3 Iterative analysis using the exponential model -- Example 4.4

Determine the current I_D and the diode voltage V_D with $V_{DD} = 5V$ and $R = 1k\Omega$. Assume that the diode has a current of 1mA at a voltage of 0.7V.

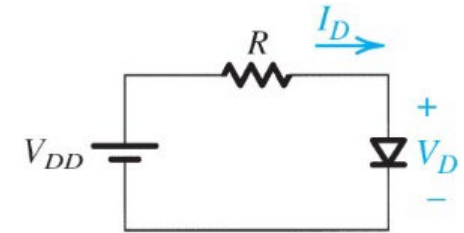
The First iteration: Assume that $V_D = 0.7V$, $I_D = \frac{V_{DD} - V_D}{R} = \frac{5V - 0.7V}{1k\Omega} = 4.3mA$

Then use the conclusion #1 to obtain a better estimate for V_D

($V_2 = 0.7V$ and $I_2 = 1mA$, $I_1 = 4.3mA$)

$$V_1 = V_2 + 0.06 \times \lg \frac{I_1}{I_2} = 0.7V + 0.06 \times \lg \frac{4.3}{1} = 0.738V$$

Thus the first iteration are $I_D = 4.3mA$ and $V_D = 0.738V$



The second iteration: Assume that $V_D = 0.738V$, $I_D = \frac{5V - 0.738V}{1k\Omega} = 4.262mA$

($V_2 = 0.738V$ and $I_2 = 4.3mA$ and $I_1 = 4.262mA$)

$$\text{Thus } V_1 = 0.738V + 0.06 \times \lg \frac{4.262}{4.3} = 0.738V$$

Thus the second iteration are $I_D = 4.262mA$ and $V_D = 0.738V$

Since these values are very close to the values obtained after the first iteration, no further iterations are necessary.

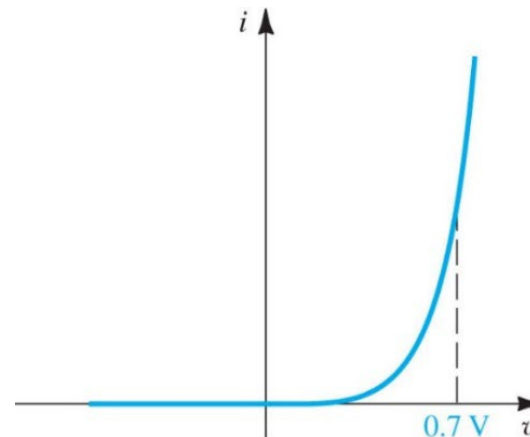
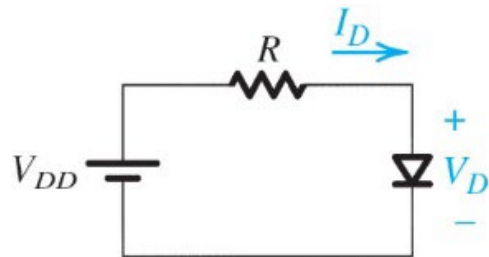
- 4.3.5 Rapid analysis using constant-voltage-drop model

$V_D = 0.7V$ (a forward-conducting diode has a voltage drop that varies in a relatively narrow range, 0.6~0.8V)

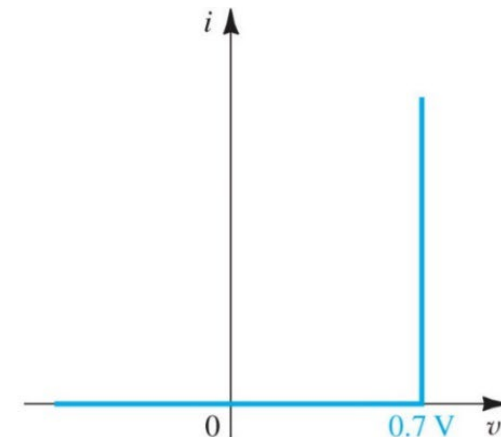
Go back to Example 4.4:

$V_{DD} = 5V$ and $R = 1k\Omega$.

$$I_D = \frac{5V - 0.7V}{1k\Omega} = 4.3mA$$

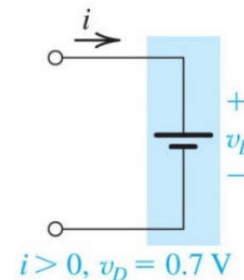


(a)



(b)

Figure 4.12 Development of the diode constant-voltage-drop model: (a) the exponential characteristic; (b) approximating the exponential characteristic by a constant voltage, usually about 0.7 V; (c) the resulting model of the forward-conducting diodes.



(c)

• 4.3.7 The small-signal model

- To determine the values of these incremental changes → small-signal model
- Linear circuit:
 - V_{DD} undergoing a small change ΔV_{DD} , Then the current I_D changes by an increment ΔI_D , and the diode voltage changes by an increment ΔV_D

$$v_D(t) = V_D + v_d(t) \text{ and } i_D(t) = I_D + i_d(t)$$

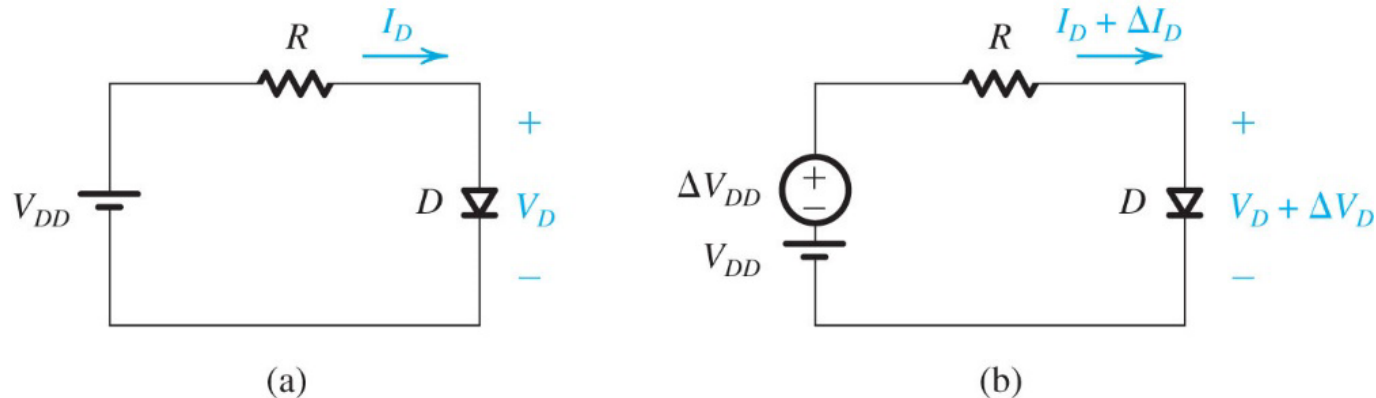


Figure 4.13 (a) A simple diode circuit; (b) the situation when V_{DD} changes by ΔV_{DD} .

- The small-signal model

Voltage across the diode - $v_D(t)$ as the sum of the dc voltage - V_D and the time-varying signal - $v_d(t)$

$$v_D(t) = V_D + v_d(t)$$

Instantaneous diode current

$$i_D(t) = I_S e^{v_D(t)/V_T} = I_S e^{(V_D + v_d)/V_T} = I_S e^{V_D/V_T} e^{v_d/V_T}$$

Since the diode current is $I_D = I_S e^{V_D/V_T}$,

$$i_D(t) = I_D e^{v_d/V_T}$$

Using **Taylor series**: $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots \frac{x^n}{n!}$ (if $x \ll 1$, $e^x \approx 1 + x$),

the small-signal model is

$$i_D(t) \approx I_D \left(1 + \frac{v_d}{V_T} \right) = I_D + \frac{I_D}{V_T} v_d \quad \text{for } \frac{v_d}{V_T} \ll 1 \Rightarrow v_{d(p)} < 5mV$$

– Diode small-signal conductance: $\frac{I_D}{V_T}$

– The inverse is the diode **small-signal resistance, or incremental resistance**, $r_d = \frac{V_T}{I_D}$

- Diode small-signal model

- dc bias point Q: characterized by the dc voltage V_D and the dc current I_D .
- Superimposed on a small signal $v_d(t)$
- In a small range, we have a linear i-v segment

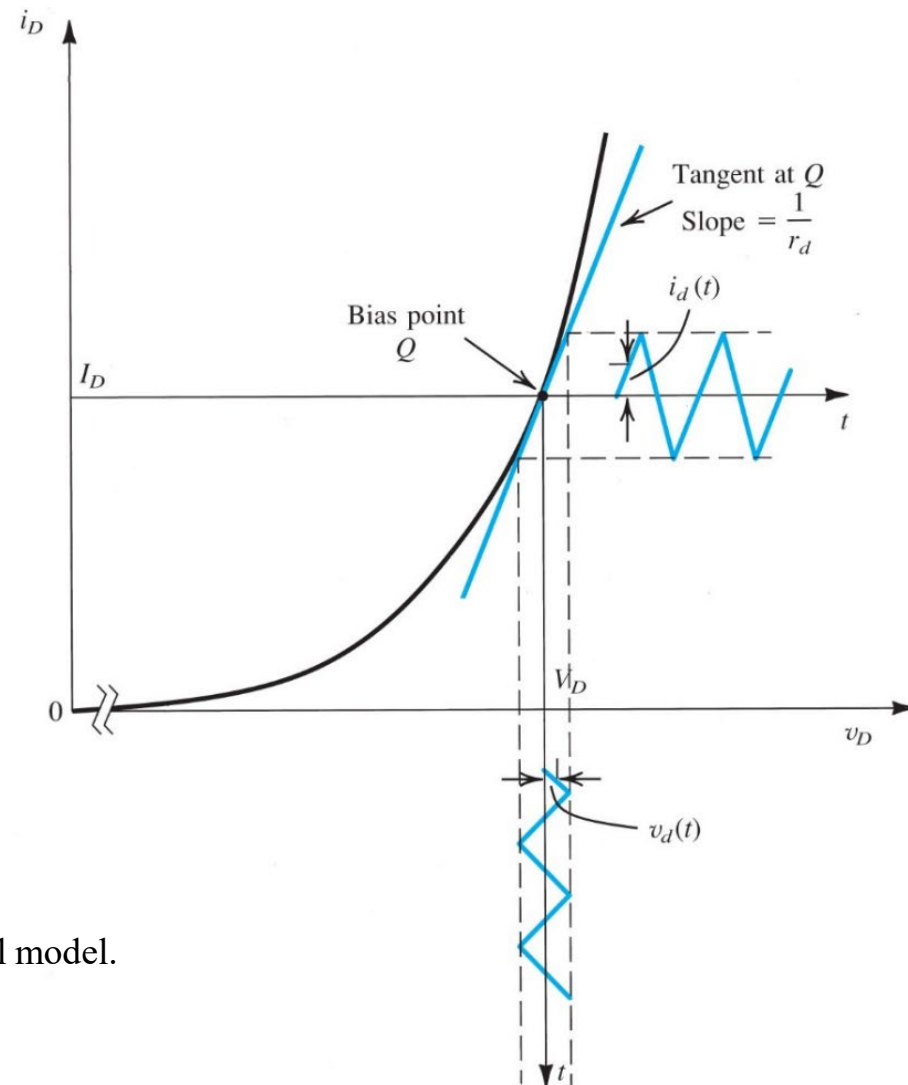


Figure 4.14 Development of the diode small-signal model.

- **Conclusion:**

- A diode circuit involves both dc and signal quantities.
- A small signal circuit can be obtained by eliminating dc sources and replacing the diode by its small-signal resistance.

- **Equivalent Models:**

- Exponential model

$$i = I_s e^{v/v_T}$$

$$v_1 - v_2 = 0.06 \times \lg \frac{I_1}{I_2}$$

- Constant-voltage-drop model ($V_D = 0.7V$)

- Small-signal model/incremental model

$$r_d = \frac{V_T}{I_d} \quad (v_{d(p)} < 5mV)$$

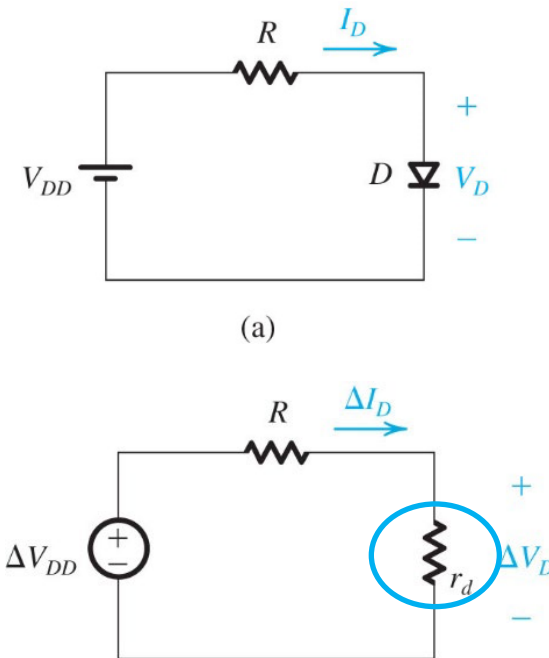


Figure 4.15 Circuit for determining the incremental quantities ΔI_D and ΔV_D for the circuit in Figure

4.13(b). Note that replacing the diode with its small-signal resistance r_d results in a linear circuit.

• Example 4.5

The power supply V^+ has a dc value of $10V$ on which is superimposed a 60-Hz sinusoid of 1-V peak amplitude. Calculate both the (a) dc voltage of the diode and (b) the amplitude of the sine-wave signal appearing across it.

Assume that the diode has 0.7-V drop at 1-mA current. $R=10k\Omega$.

(a) Consider the dc quantities only, we assume $V_D = 0.7V$. (iterative analysis)

$$I_D = \frac{10V - 0.7V}{R} = 0.93mA, V_1 = 0.7 + 0.06 \times \lg \frac{0.93}{1}, V_1 = 0.698V$$

Very close to 1mA and 0.7V

(b) Then, assume $v_d(\text{peak}) < 5mV$

the diode incremental resistance

$$r_d = \frac{V_T}{I_d} = \frac{25mV}{0.93mA} = 26.9\Omega$$

$$v_d(\text{peak}) = \frac{r_d}{r_d + R} \times v_s(\text{peak}) = 2.68mV (< 5mV)$$

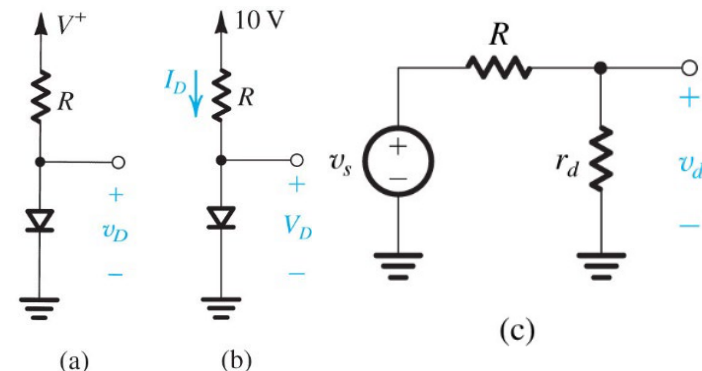


Figure 4.16 (a) Circuit for Example 4.5. (b) Circuit for calculating the dc operating point. (c) Small-signal equivalent circuit.

• Example 4.6: voltage regulator

Goal: Provide a constant voltage of 2.1V. Calculate the **change in output voltage** caused by (a) a $\pm 10\%$ change in the power-supply voltage.

(b) connection of a 1-k Ω load resistance.

$$(a) I_D = \frac{10V - 2.1V}{1k\Omega} = 7.9mA,$$

Assume small-signal, thus each diode has a resistance

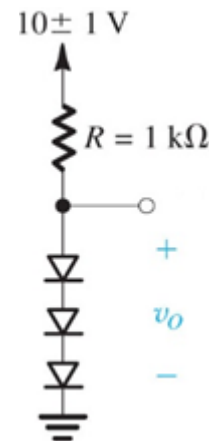
$$r_d = \frac{V_T}{I_d} = \frac{25mV}{7.9mA} = 3.2\Omega \quad \text{so } 3r_d = 9.6\Omega$$

The peak change in output will be

$$\Delta v_o = \frac{9.6\Omega}{9.6\Omega + 1k\Omega} \times \pm 1V = \pm 9.5mV$$

Implies a change of **$\pm 3.2mV$ per diode ($< 5mV$)** -> justified

Figure 4.17 (a) Circuit for Example 4.6.



• Example 4.6 Cont.

Goal: Provide a constant voltage of 2.1V. Calculate the **change in voltage** caused by (a) a $\pm 10\%$ change in the power-supply voltage.

(b) connection of a 1-k Ω load resistance.

(b) As (a), $I_D = 7.9\text{mA}$, and each diode has a resistance $r_d = 3.2\Omega$, so $3r_d = 9.6\Omega$

The current through R_L is $\frac{2.1\text{V}}{1\text{k}\Omega} = 2.1\text{mA}$,

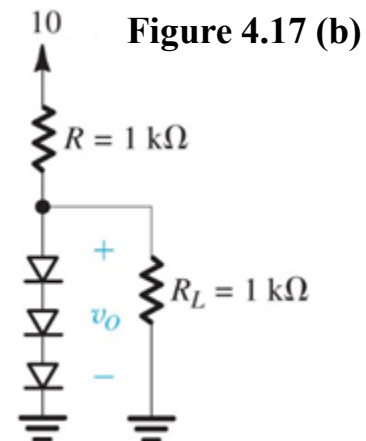
Assume small-signal, A voltage decrease on diode $\Delta v_o = -2.1\text{mA} \times 9.6\Omega = -20\text{mV}$

Implies a change of **$\pm 6.7\text{mV}$ per diode ($> 5\text{mV}$) $\rightarrow$ not justified**

Using exponential model :

Current reduces from $i_2 = 7.9\text{mA}$ to $i_1 = (7.9\text{mA} - 2.1\text{mA}) = 5.8\text{mA}$

$$\Delta v_d = 0.06 \times \lg \frac{5.8}{7.9} = -7.8\text{mV} \text{ and } \Delta v_o = 3\Delta v_d = -23\text{mV}$$



4.5 Rectifier circuits

- AC power supply: 120V(rms) 60-Hz
- Transformer:
 - Primary winding (N_1 turns) and Secondary winding (N_2 turns)
 - $v_S: 120 \times \frac{N_2}{N_1} V(rms)$ e.g. 8-V (rms) v_S , 15:1 turns ratio
- Diode rectifier: sine-wave $\rightarrow$ unipolar output
- Filter: reduce the variations of the rectifier output
- Voltage regulator: reduce ripple (zener shunt regulator)

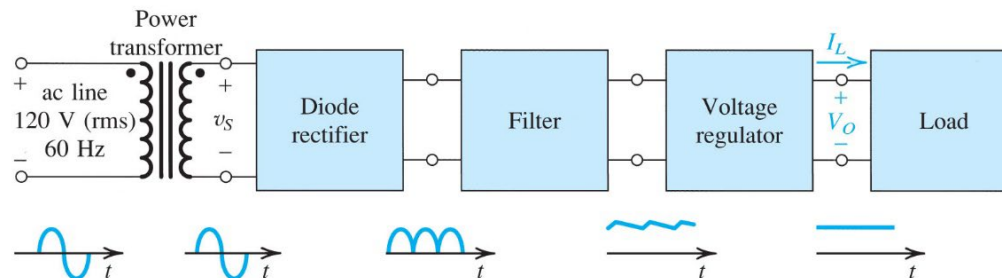


Figure 4.22 Block diagram of a dc power supply.

- 4.5.1 The half-wave rectifier

$$v_o = 0, v_s < V_D$$

$$v_o = v_s - V_D, v_s \geq V_D \quad V_D \text{ is around } 0.7\text{V}$$

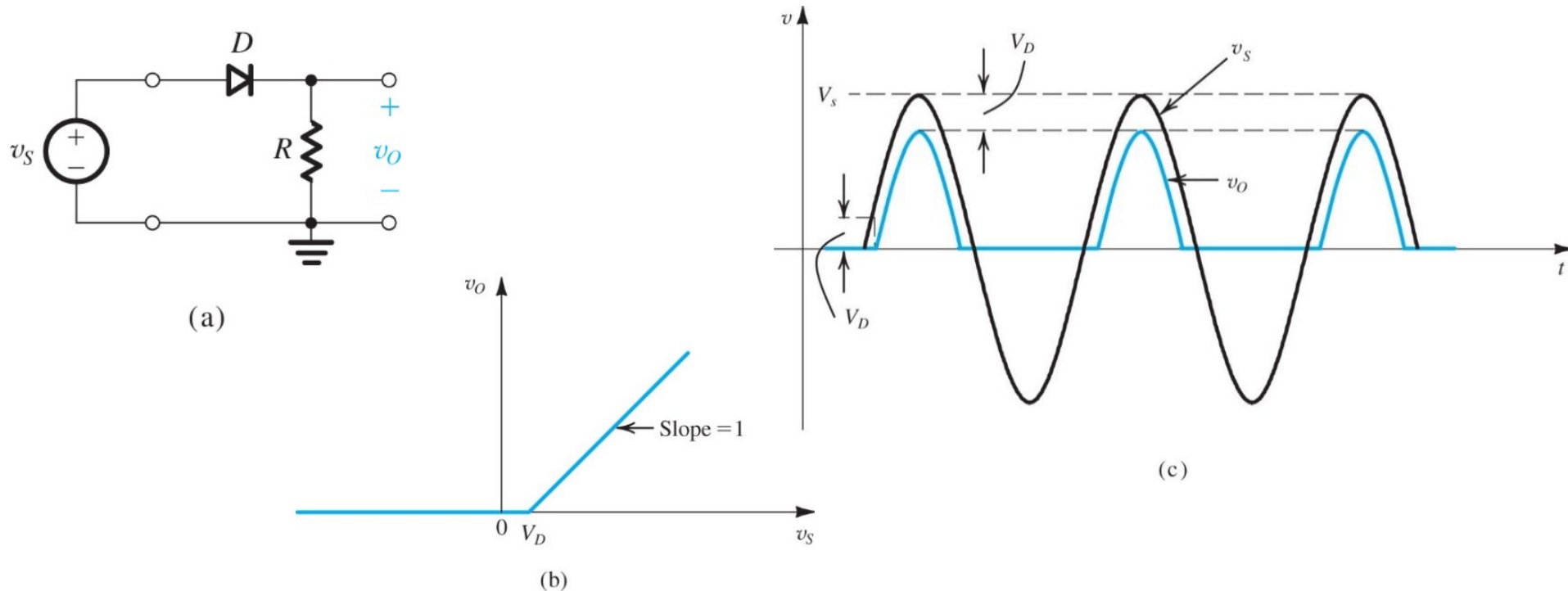


Figure 4.23 (a) Half-wave rectifier. (b) Transfer characteristic of the rectifier circuit. (c) Input and output waveforms.

• 4.5.2 The full-wave rectifier

- **Center-tap** the transformer secondary winding

- v_S positive half: D1 conducts, D2 reverse bias

the circuit behaves like a half-wave rectifier

- v_S negative half: D1 cut off, D2 conducts

the circuit behaves again like a half-wave rectifier

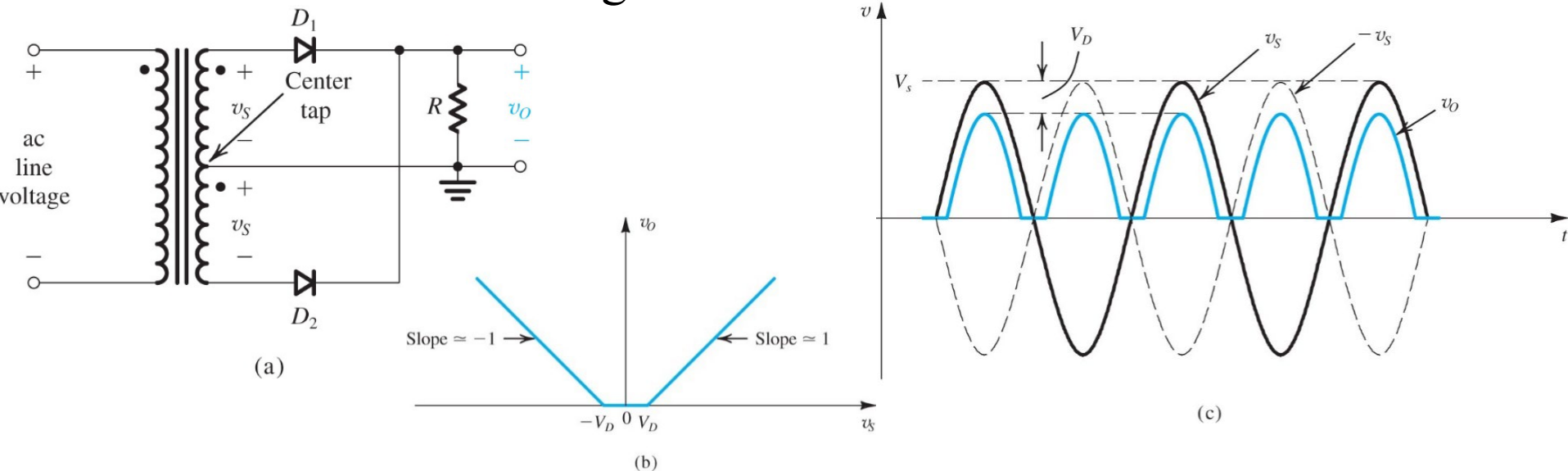


Figure 4.24 Full-wave rectifier utilizing a transformer with a center-tapped secondary winding: (a) circuit; (b) transfer characteristic assuming a constant-voltage-drop model for the diodes; (c) input and output waveforms.

• 4.5.3 The bridge rectifier

- v_S positive: D1 and D2 conduct, D3 and D4 cutoff
- v_S negative : D1 and D2 cut off, D3 and D4 conduct
- Advantages: Don't require a center-tap transformer
- Disadvantages: 1) four diodes 2) double voltage drop on output

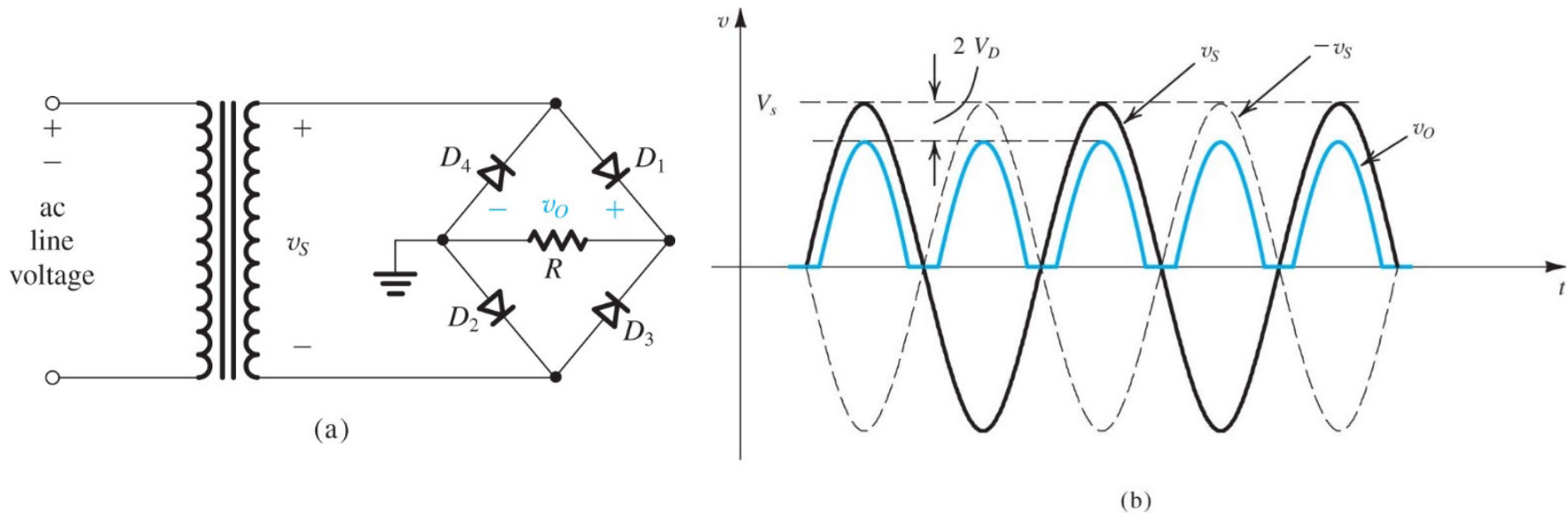
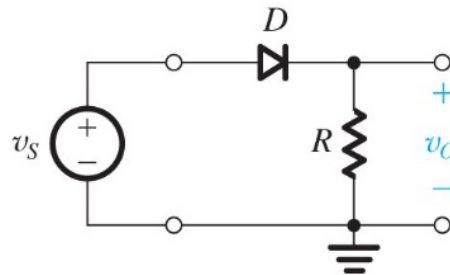


Figure 4.25 The bridge rectifier: (a) circuit; (b) input and output waveforms.

• Experimental 5.1

Do the computer **simulation** and **set up** the following circuits.
Set up the following circuit with diode (1N4148).



$v_{in}(t) = V_p \sin(\omega t)$ volts, $V_p = 5V$, $f = 60Hz$. $R = 1K\Omega$.

Measure and **graph** $v_o(t)$. Briefly explain and comment your results.

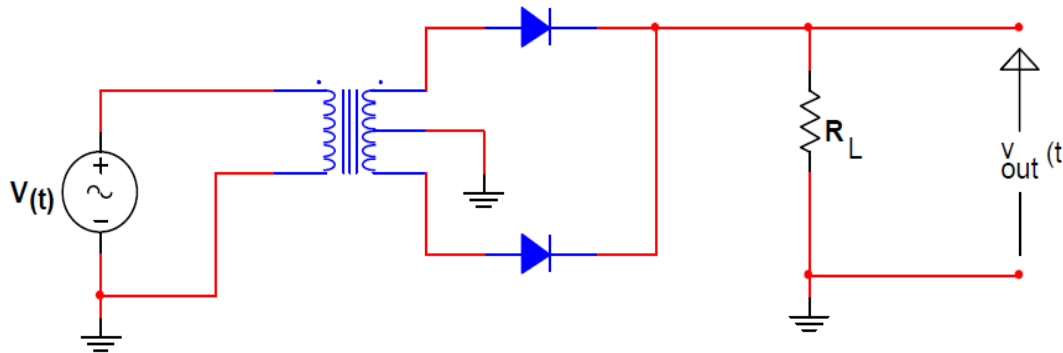
Experimental results:

$v_{in}(t)$	R	v_{omax}	v_{omin}
$V_p = 5V, f = 60Hz$	$1K\Omega$		

• Experimental 5.2

Do the computer **simulation** of the following circuits.

Set up the following circuit with the transformer and diode (1N4148).



Repeat the procedure 1.

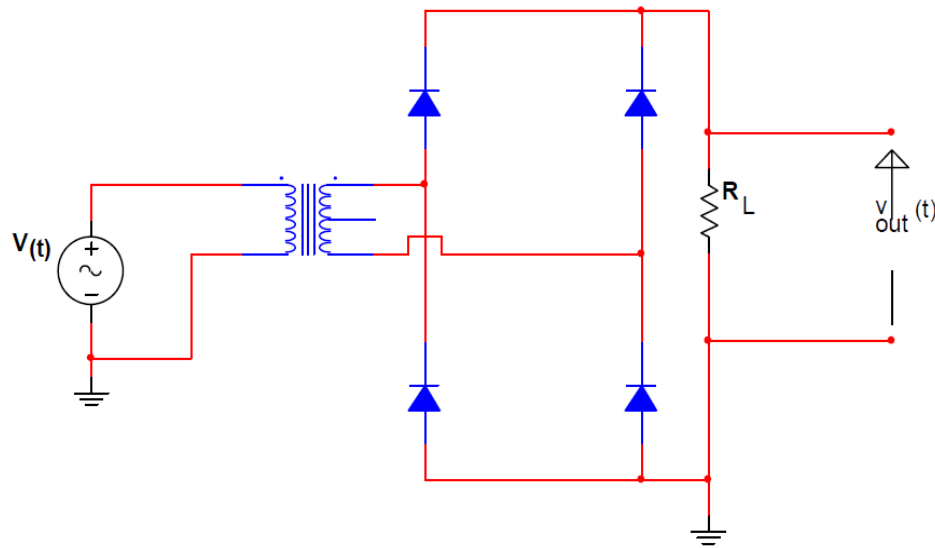
Measure and graph $v_o(t)$. Briefly explain and comment your results.

$v_{in}(t)$	R_L	v_{omax}	v_{omin}
$V_p = 40V, f = 60Hz$	$10k\Omega$		

• Experimental 5.3

Do the computer **simulation** of the following circuits.

Set up the following circuit with the transformer and diode (1N4148).



$v_{in}(t)$	R_L	v_{omax}	v_{omin}
$V_p = 40V, f = 60Hz$	$10k\Omega$		

$$v_{in}(t) = V_p \sin(\omega t) \text{ volts.}$$

Measure and graph $v_o(t)$. Briefly explain and comment your results.

4.6 Limiting and clamping circuits

- 4.6.1 Limiter/clipping circuit: $\frac{L_-}{K} \leq v_I \leq \frac{L_+}{K}$
 - Clippers: the positive and/or negative peaks are clipped off
 - Used in FM transmitters where noise peaks are limited to a particular value so that excessive peaks are removed from them.
 - Used to put off the voltage beyond the preset value without disturbing the remaining part of the input waveform.

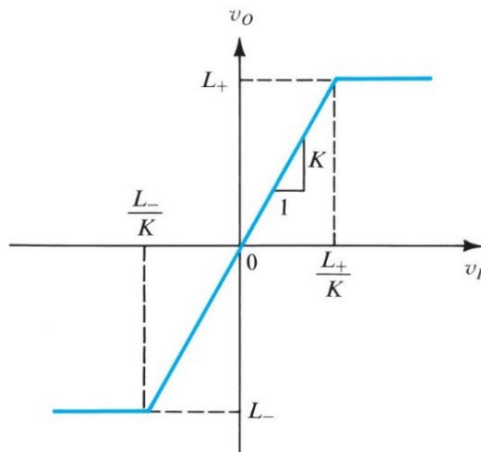


Figure 4.30 General transfer characteristic for a limiter circuit.

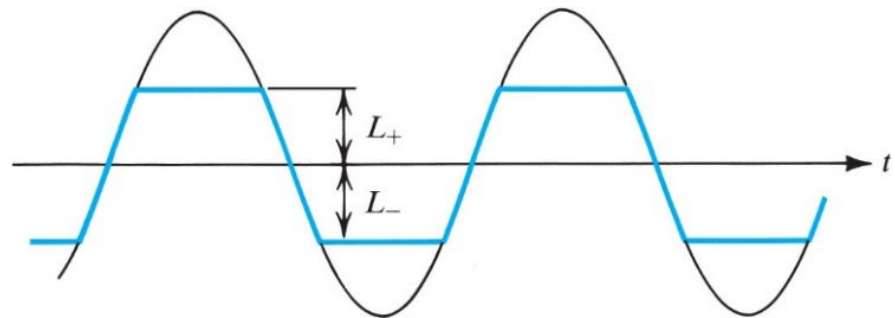


Figure 4.31 Applying a sine wave to a limiter can result in clipping off its two peaks.

- Fig. 4.33 (a)
 - $v_I < 0.5V$, diode is cut off, $v_O = v_I$
 - $v_I \geq 0.5V$, diode conducts, v_O will be saturated at one diode drop (0.7V)
- Fig. 4.33 (b)
 - $v_I > -0.5V$, diode is cut off, $v_O = v_I$
 - $v_I \leq -0.5V$, diode conducts, v_O will be saturated at one diode drop (-0.7V)

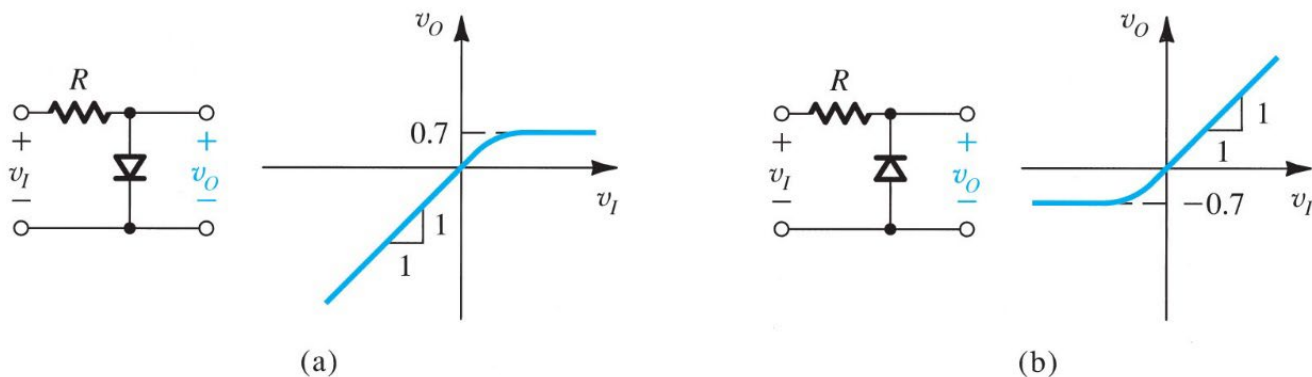
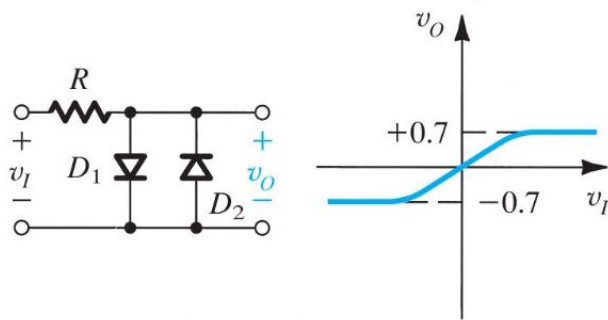
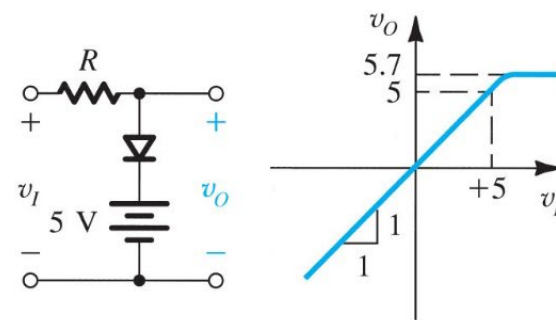


Figure 4.33 A variety of basic limiting circuits. (using the constant-voltage-drop (V_D) diode model)

- Fig. 4.33 (c): double limiting
 - $-0.5V \leq v_I \leq 0.5V$, both diodes are cut off, $v_o = v_I$
 - $v_I > 0.5V$, D1 conducts, v_o will be limited to $+0.7V$
 - $v_I < -0.5V$, D2 conducts, v_o will be limited to $-0.7V$
- Fig. 4.33 (d): threshold level control
 - $v_I < 5.5V$, diode is cut off, $v_o = v_I$
 - $v_I \geq 5.5V$, diode conducts, v_o will be saturated at $5+0.7=5.7V$



(c)



(d)

Figure 4.33 A variety of basic limiting circuits. (using the constant-voltage-drop (V_D) diode model)

• Exercise 4.26

Assume the diodes to be ideal (no 0.7V drop). Describe the transfer characteristic of the circuit.

When $-5V < v_I < 5V$, D_1 and D_2 cut off. $v_o = v_I$

When $v_I \geq 5V$, D_2 conducts, D_1 cut off

$$\text{KVL: } v_I - 5 - I \times 20 = 0 \Rightarrow I = \frac{v_I - 5}{20}$$

$$v_o = v_I - I \times 10 = \frac{v_I}{2} + 2.5V$$

When $v_I \leq -5V$, D_1 conducts, D_2 cut off

$$\text{KVL: } v_I + 5 - I \times 20 = 0 \Rightarrow I = \frac{v_I + 5}{20}$$

$$v_o = v_I - I \times 10 = \frac{v_I}{2} - 2.5V$$

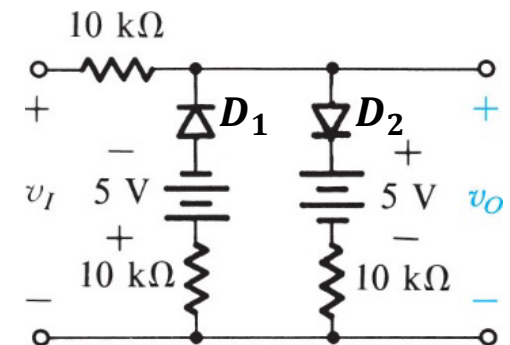


Figure E4.26

• 4.6.2 Clamping circuit

- Used to shift/alter either positive or negative peak of an input signal to a desired level, also called as level shifter or DC restorer. These clamping circuits can be positive or negative depends on the diode configuration.

- Negative part: diode conducts, capacitor is charged.

Ideal model: $v_I - v_c = 0$. $v_c = v_I = -6V$

- Positive part: diode cut off

Ideal model: $v_o = v_I - v_c = 4 + 6 = 10V$

- The lowest peak **clamped** to 0V -> clamped circuit

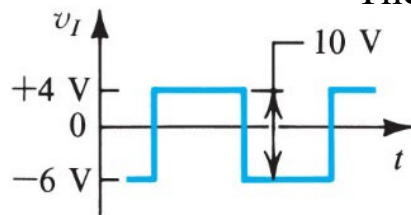
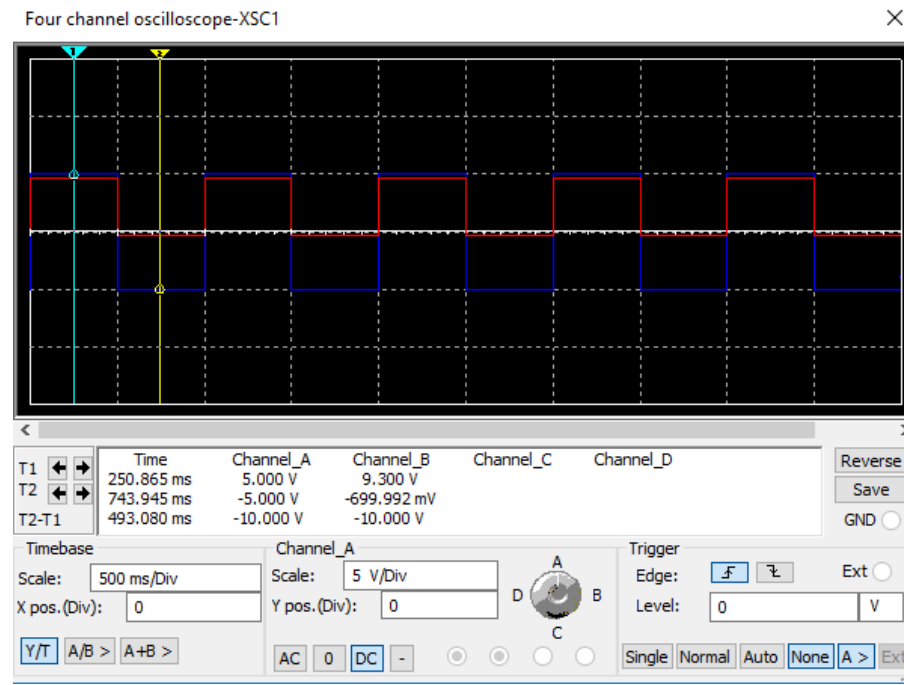
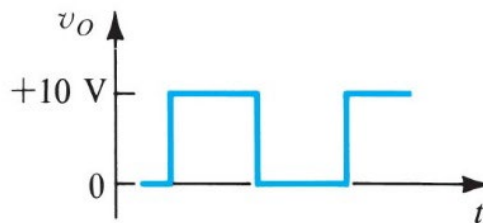
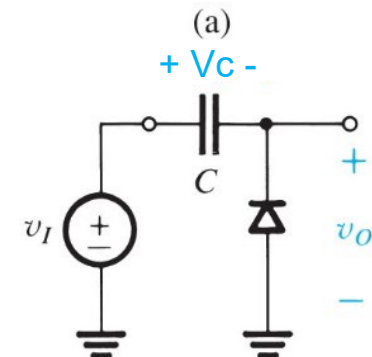


Figure 4.34 The clamped capacitor or dc restorer with a square-wave input and no load.



- The clamped capacitor with a load resistance R

- t_0 to t_1 : discharge with time constant CR
- t_1 : v_I decreases by V_a and v_o follows; diode conducts and charge the capacitor
- t_2 : v_I rises by V_a and v_o follows
- The charge lost by the capacitor during the interval t_0 to t_1 is recovered during the interval t_1 to t_2

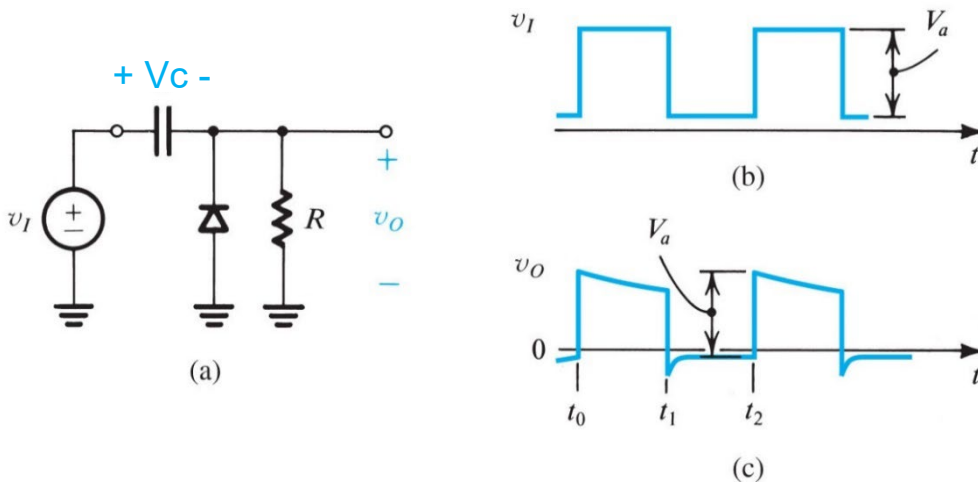
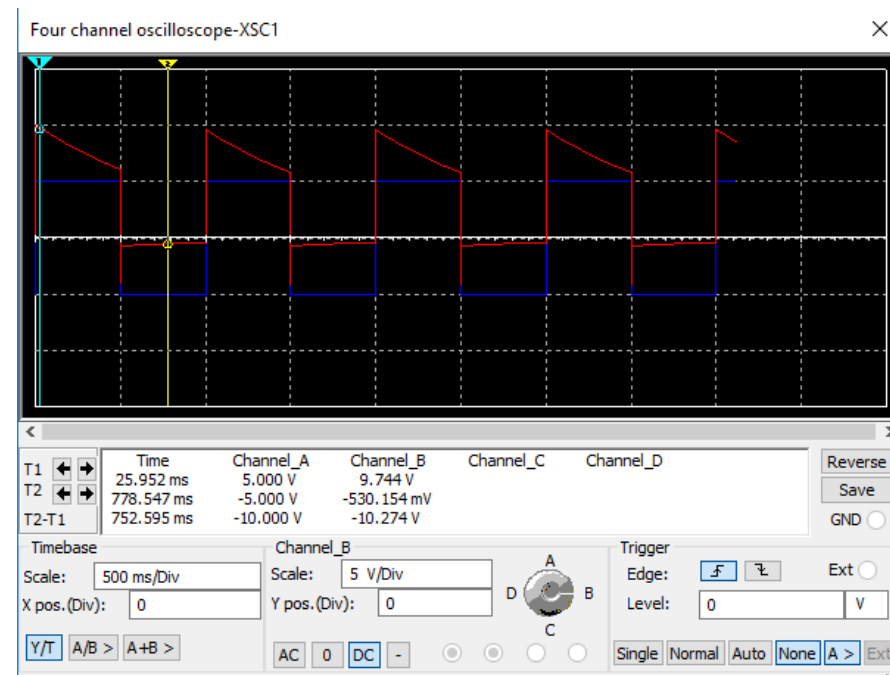


Figure 4.35 The clamped capacitor with a load resistance R .



• 4.6.3 The voltage doubler

- Positive v_I : D1 conducts, D2 cutoff

Ideal model, $v_I - v_{C1} = 0$, thus $v_{C1} = v_I|_{max} = V_P$

- Negative v_I : D2 conducts, D1 cutoff

Ideal model, $v_{C2} + V_P - v_I = 0$,

thus $v_o = v_{C2} = v_I|_{max} - V_P = -2V_P$

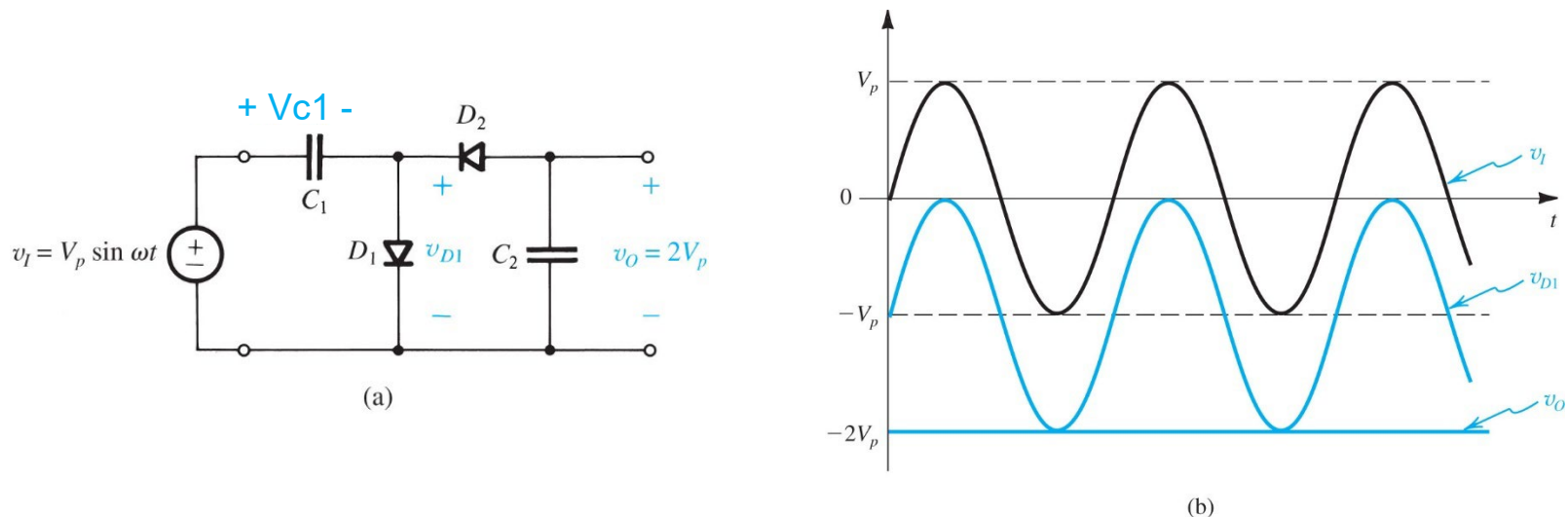
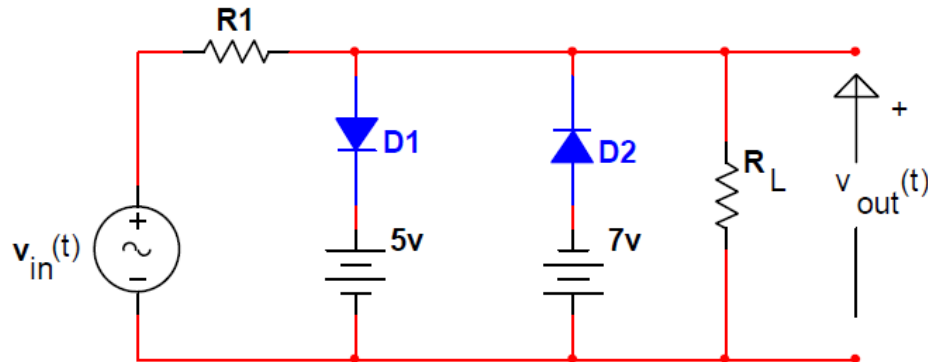


Figure 4.36 Voltage doubler: (a) circuit; (b) waveforms of the input voltage, the voltage across D1, and the output voltage $v_o = -2V_p$.

- Experiment 6.1

Set up the following circuit with diode (**1N4148**).



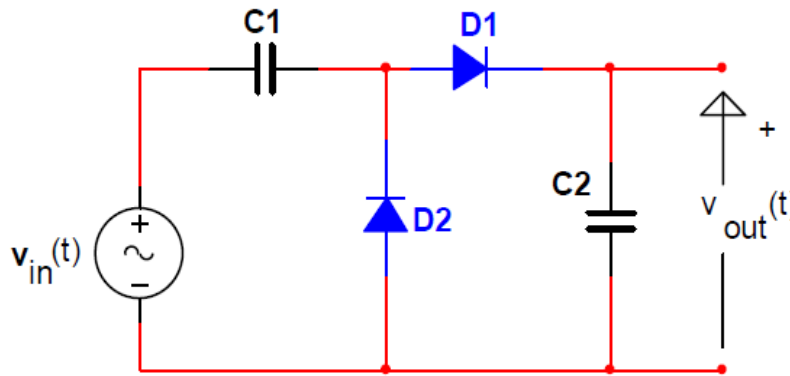
$v_{in}(t) = V_p \sin(\omega t)$ volts.

Measure and **graph** $v_o(t)$. Briefly explain and comment your results.

$v_{in}(t)$	R_1	R_L	$v_{o\max}$	$v_{o\min}$
$V_p = 10V, f = 60Hz$	100Ω	$10k\Omega$		

- Experiment 6.2

Set up the following circuit with diode (**1N4148**).



$$v_{in}(t) = V_p \sin(\omega t) \text{ volts.}$$

Measure and **graph** $v_o(t)$. Briefly explain and comment your results.

$v_{in}(t)$	C_1	C_2	v_{omax}	v_{omin}
$V_p = 10V, f = 60Hz$	$0.1\mu F$	$0.1\mu F$		

HW3-4

- Problems
 - PP.236, 4.37 – Modeling the diode forward characteristic – the exponential model
 - PP.238, 4.56 – Modeling the diode forward characteristic – the small-signal model
 - PP.239, 4.70 – Rectifier circuits
 - PP.239, 4.96 – Limiting and Clamping Circuits
- Submission requirement:
 - Add the cover page!!!
 - [Print the HW3-4.pdf out and answer all the questions \(download on the blackboard\)](#)
- HW3-4 Due: TBA (Late assignments: 40% deduction)